Explainable AI for Academic Reviews

A Multi-Task Intelligence Engine for Academic Reviews

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DATA-441 | Final Project Presentation

Rate My Professor (RMP)

The Platform

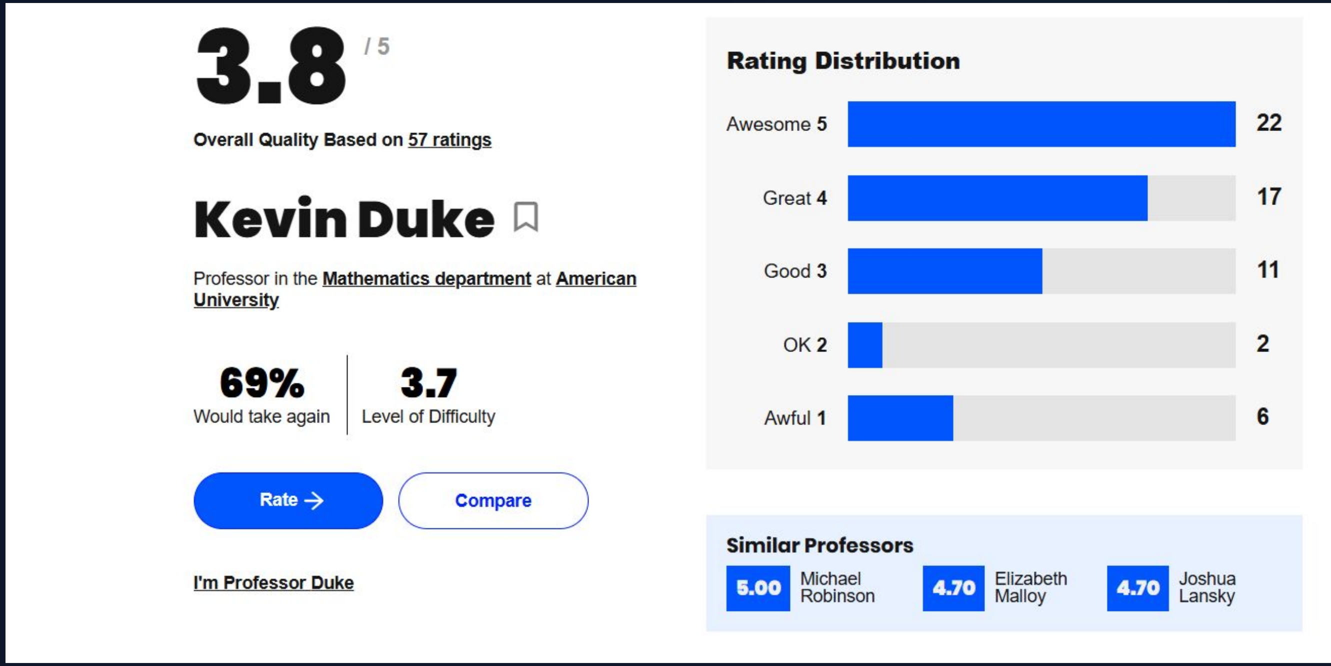
The de facto standard for student course registration, hosting millions of crowdsourced reviews.

The Data Structure

- **Quantitative:** 1-5 scales for Quality and Difficulty.
- **Categorical:** Pre-selected attribute tags.
- **Unstructured:** Open-ended text reviews.

The Impact

A primary decision-making tool dictating university enrollment.

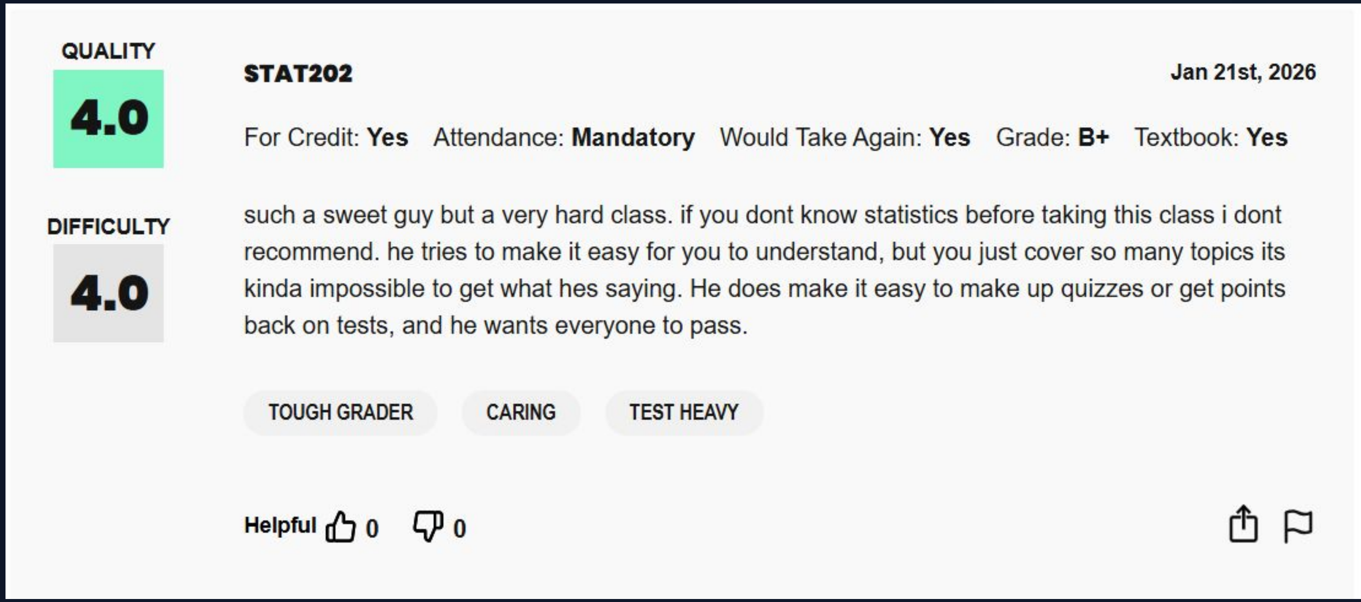


The "Black Box" Problem

Ambiguity

-When a student sees a "4 Difficulty" on RMP, it's a black box. Why is it hard? Is the grading unfair, or is there just a lot of reading?

-Furthermore, sorting bias dictates perception. Students often only read the first few reviews, which might not represent the whole professor profile.



Goals:

Allow RMP viewers to understand the potential language that is driving students reviews.

Bring Amazon-style AI Consensus Summaries to academia.

The Challenge of Student Language

Informal Structures

Reviews are essentially "Academic Tweets." They lack formal grammar and follow unstructured semantic flows.

Heavy Slang & Sarcasm

Students use niche internet slang, emojis, and sarcasm to convey frustration or praise, which is difficult for standard algorithms to parse.

NLP Objective

Our objective was to build a system that parses all reviews simultaneously to decode these signals into a structured profile.

Guess the Score!

I’ll show a review, you guess the score

Guess the Score!

Difficulty Score: ?

“He wants you to get an A more than you do! Greatest guy out there. He is extremely accessible and makes class extremely funny and engaging. Easy to learn and you don't notice how much you have retained until later. Strict on attendance and participation, other than that will be a very chill class. If it's an 8am take him over any other professor.”

Guess the Score!

Difficulty Score: 1

Guess the Score!

Quality Score: ?

“She is not a good professor, overall the class is very lecture heavy and quite boring. She's been very condescing and leaves vague comments on the only two assignments we have graded despite having weekly responses to the readings. The class isn't difficult but I feel like she makes interesting topics dreadfully boring.”

Guess the Score!

Quality Score: 1

Guess the Score!

Difficulty Score: ?

"She's an ANGEL. This class is dead easy, just stay on top of readings... She is very knowledgeable and she can be intimidating, but she's awesome overall -- you will be rewarded for your hard work."

Guess the Score!

Difficulty Score: 2

Guess the Score!

Quality Score: ?

"She is not for coasters. She is tough and is not afraid to tell you how wrong you are... If you genuinely want to learn something, she's amazing. If you're just trying to check of a requirement, avoid at all costs."

Guess the Score!

Quality Score: 5

Guess the Score!

Difficulty Score: ?

"If you have the chance to take a class with him, you have to. Even though I am not a morning person I took his 8am, and I don't regret it at all. He is there to help every student succeed and if you follow directions you will get an A for sure. He also brings in amazing guest speakers. Overall the best!"

Guess the Score!

Difficulty Score: 5

Data Collection

- **REST API Deprecation:** Rate My Professor recently shut down its public database, making standard scraping methods obsolete.
- **Modern Security:** The platform now uses a heavily protected GraphQL backend guarded by Cloudflare bot-detection.
- **Legacy Failure:** Open-source tools from GitHub (like tisuela/ratemyprof-api) fail instantly against current anti-bot handshakes.
- **The Solution:** Modify existing scraping method to work with modern security. Utilized **curl_cffi** library to impersonate Chrome browser TLS communication.

```
response = requests.post(
    "https://www.ratemyprofessors.com/graphql",
    headers=self.headers, json={"query": query, "variables": variables}, impersonate="chrome"
)
```

Dual-Pipeline Architecture

Branch A: Predictive (CNN)

Aggressive regex cleaning, lowercasing, and padding to 150 tokens.

Branch B: Generative (LLM)

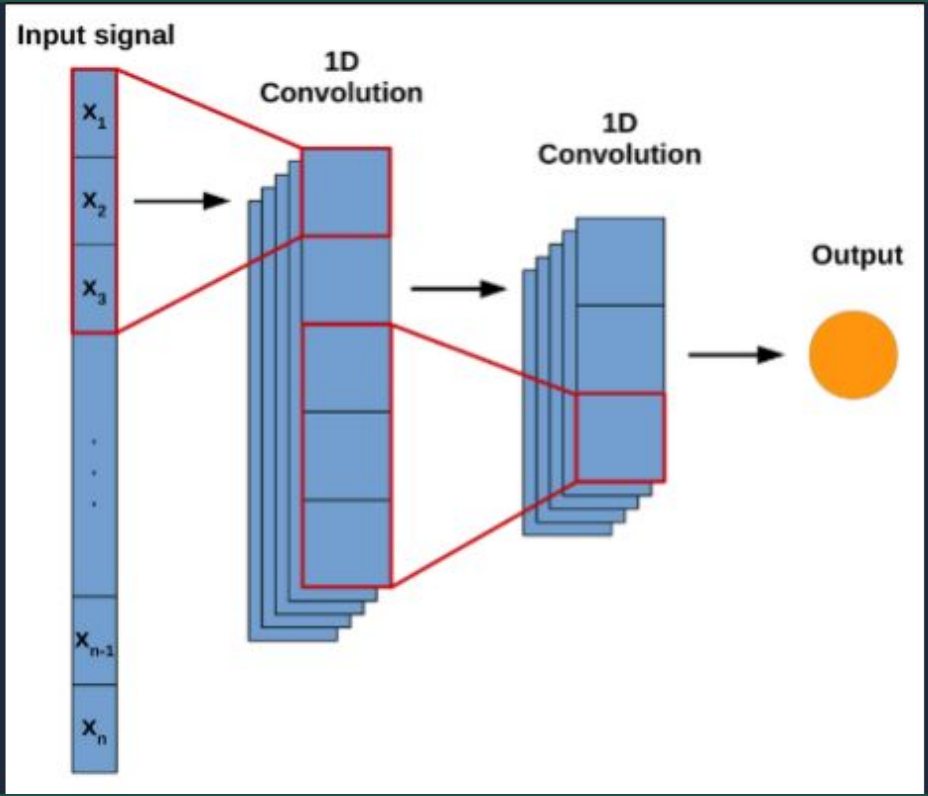
Qualitative sentiment preservation. Retains punctuation, capitalization, and emotional semantic weight.

	A	B	C	D	E	F	G	H	I
1	professor_	professor_	course	review_tex	grade	difficulty	quality	would_take	tags
2	1674106	Mary Ellen	HX305	She's really	A	2	5	Yes	Caring, Lec
3	1674106	Mary Ellen	AMST240	She is not a	Not sure ye	2	1	N/A	Tough grad
4	1674106	Mary Ellen	AMST240	She is AWE	Not sure ye	2	5	N/A	Tough grad
5	1674106	Mary Ellen	AMST240	She is the b	A-	2	5	Yes	Amazing le
6	1674106	Mary Ellen	AMST240	She is the b	A+	2	5	Yes	Amazing le
7	1674106	Mary Ellen	AMST375	Super carir	A	3	5	Yes	Get ready t
8	1674106	Mary Ellen	AMST375	Caring prof	A	3	4	Yes	Participati
9	1674106	Mary Ellen	AMST240	Very readir	B	3	4	Yes	Get ready t
10	1674106	Mary Ellen	WGSS350	I took Dr. C	Not sure ye	4	5	Yes	Get ready t
11	1674106	Mary Ellen	AMST375	Couldn't ev	Not sure ye	4	1	N/A	Tough grad
12	1674106	Mary Ellen	AMST140	I actually h	A+	3	3	N/A	Participati
13	1674106	Mary Ellen	AMST140	A very critic	Not sure ye	3	4	Yes	Tough grad
14	1674106	Mary Ellen	WGSS350	I thought I w	B+	3	1	N/A	Get ready t
15	1674106	Mary Ellen	AMST240	I loved her! One of the		2	5	Yes	Get ready t
16	1674106	Mary Ellen	AMST240	I love Dr. C	A	3	5	Yes	Gives good
17	1674106	Mary Ellen	AMST320	Honestly o	A	3	5	Yes	Gives good
18	1674106	Mary Ellen	AMST240	Dr. Curtin i	A	2	5	Yes	Inspiration
19	1674106	Mary Ellen	AMST240	She is a very sweet and		4	5	Yes	Gives good
20	1674106	Mary Ellen	AMST140	She's so sw	A	2	3	Yes	Get ready t
21	1674106	Mary Ellen	AMST240	she's very r	A-	3	3	No	Get ready t
22	1674106	Mary Ellen	HX305	The best te	A-	3	5	Yes	Get ready t
23	1674106	Mary Ellen	AMST240	Mary Ellen	A-	3	4	Yes	Respected
24	1674106	Mary Ellen	WGSS150	this class has only pap		3	5	Yes	Gives good
25	1674106	Mary Ellen	WGSS150	Professor (	A	4	5	Yes	Get ready t
26	1674106	Mary Ellen	AMST240	Professor (	A	4	2	No	Tough Grad
27	1674106	Mary Ellen	AMST240	Prof. Curtir	A	2	3	Yes	Participati
28	1674106	Mary Ellen	HIST220	Curtin is a	A+	3	4	Yes	
29	1674106	Mary Ellen	HIST215	Very helpful with work		3	5	N/A	

Branch A: Multi-Task 1D CNN

N-Gram Feature Detection

- **Dense Text Embeddings:** Maps informal student slang and academic jargon into a structured semantic vector space.
- **N-Gram Feature Detection:** Unlike Bag-of-Words, a 1D convolution slides a "window" across the text to capture contextual phrases (e.g., "heavy reading load") rather than isolated words.
- **Global Max Pooling:** Acts as a filter to extract only the absolute strongest semantic signals from the entire review, ignoring filler words.
- **Multi-Task Regression Heads:** The shared network splits into two simultaneous outputs. By predicting Quality and Difficulty together, the model learns the nuanced correlation between the two.



Quantitative Accuracy

0.42

Quality MAE

0.54

Difficulty MAE

Predictive Power

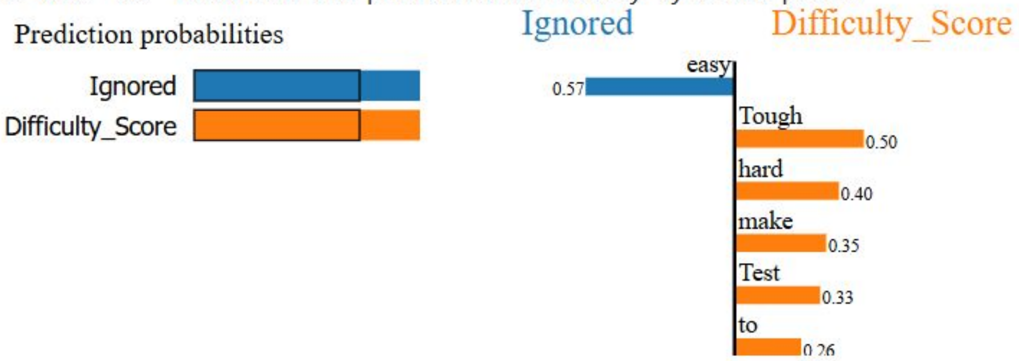
On a 5-point scale, our model predicts numerical ratings with an error of just roughly half a point. This validates that the CNN successfully mapped informal language to teacher scores.

Displaying our Model with LIME (Difficulty)

```
Original Text: such a sweet guy but a very hard class. if you dont know statistics before taking this class i dont recommend. he tries to make it easy for you to understand,
Actual Student Difficulty Rating: 4

--- LIME EXPLANATION ---
Model Predicted Difficulty: 4.07

Top words driving this prediction:
The word 'easy' DECREASED the predicted difficulty by 0.572 points
The word 'Tough' INCREASED the predicted difficulty by 0.496 points
The word 'hard' INCREASED the predicted difficulty by 0.401 points
The word 'make' INCREASED the predicted difficulty by 0.352 points
The word 'Test' INCREASED the predicted difficulty by 0.328 points
The word 'to' INCREASED the predicted difficulty by 0.255 points
```



Text with highlighted words

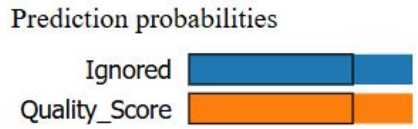
such a sweet guy but a very hard class. if you dont know statistics before taking this class i dont recommend. he tries to make it easy for you to understand, but you just cover so many topics its kinda impossible to get what hes saying. He does make it easy to make up quizzes or get points back on tests, and he wants everyone to pass. Tough grader, Caring, Test heavy

Displaying our Model with LIME (Quality)

```
Original Text: such a sweet guy but a very hard class. if you dont know statistics before taking this class i dont recommend. he tries to make it easy for you to understand,
Actual Student Quality Rating: 4

--- LIME EXPLANATION (QUALITY) ---
Model Predicted Quality: 3.47

Top words driving this prediction:
The word 'Caring' INCREASED the predicted quality by 0.872 points
The word 'Tough' DECREASED the predicted quality by 0.418 points
The word 'heavy' DECREASED the predicted quality by 0.329 points
The word 'so' INCREASED the predicted quality by 0.144 points
The word 'dont' DECREASED the predicted quality by 0.141 points
The word 'easy' INCREASED the predicted quality by 0.132 points
```



Text with highlighted words

such a sweet guy but a very hard class. if you dont know statistics before taking this class i dont recommend. he tries to make it easy for you to understand, but you just cover so many topics its kinda impossible to get what hes saying. He does make it easy to make up quizzes or get points back on tests, and he wants everyone to pass. Tough grader, Caring, Test heavy

Branch B: Generative Synthesis (LLM)

Goal:

Generate an Amazon-style "Customers Say" summary to provide an immediate qualitative overview.

System Pipeline:

Input: Query a specific professor by name.

Processing: The system aggregates all historical text reviews and feeds them into the Gemini 1.5 API via a massive context window.

Output: The LLM returns a synthesized consensus paragraph, automatically isolating and highlighting one notable "Outlier Opinion."

Generative Synthesis (LLM)

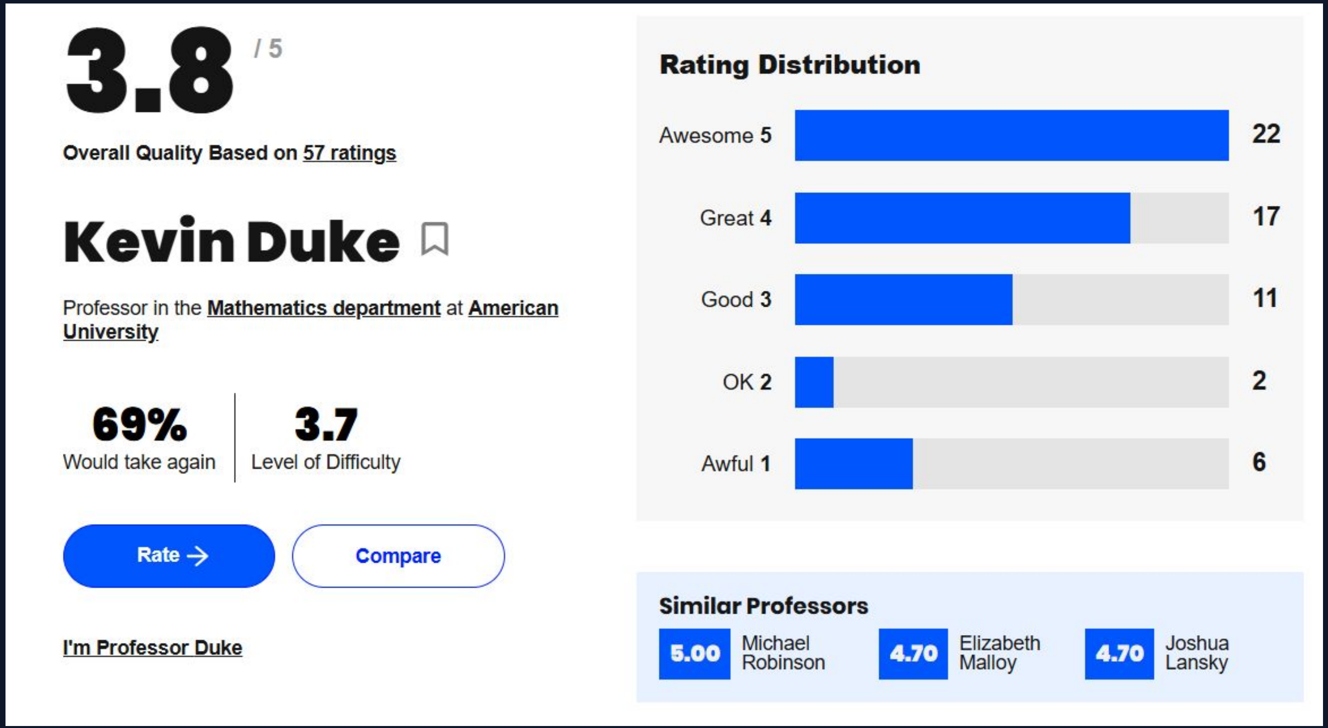
Prompt:

```
prompt = f"""
You are an expert academic advisor analyzing student feedback.
Below are all {num_reviews} of the Rate My Professor reviews for Professor {prof_name}.

Your task:
1. Read through all the reviews carefully.
2. Write a concise, 3-sentence consensus summarizing the general student sentiment.
3. Conclude with exactly 1 distinct sentence starting with "Outlier Opinion:" that
   highlights a notable dissenting view or unique experience found in the reviews.

Here are the reviews to analyze:
{combined_text}
"""
```

Branch B: Generative Synthesis (LLM): Example



Professor Duke is widely regarded as a kind, caring, and passionate instructor who genuinely wants his students to succeed, consistently offering extensive support during office hours. However, his in-class lectures are often fast-paced, theoretical, and can be confusing, particularly for students without prior knowledge in introductory math courses like statistics or pre-calculus. Despite the challenging material and teaching style, he provides numerous opportunities for extra credit, make-up work, and grade recovery, making it possible for dedicated students to pass and even excel.

Outlier Opinion: One student had a uniquely negative experience in his "Great Ideas" course, describing him as unorganized, unhelpful, and unclear, directly contradicting the overwhelmingly positive sentiment found in reviews for his math classes.

Limitations & Challenges

Hardware Constraints

Initial attempts at local Gemma-4 deployment failed due to Out-Of-Memory (OOM) errors on free-tier GPUs in Google Colab.

Fix: Pivot to Cloud API infrastructure.

API Rate Limits

Encountered strict 429 and 503 errors during massive iteration on thousands of reviews.

Fix: Engineered automated exponential backoff retry loops.

Token Context Window

Gemma 4 was limited to a few hundred thousand tokens, insufficient for professors with high review volumes.

Fix: Switched to Gemini 1.5 API with a million-token context window.

Conclusion & Future Scope

Chrome Extension Deployment

To avoid RMP legal/scraping conflicts, the next step is a client-side Chrome Extension.

The system would perform inference in real-time, reading reviews in the student's browser and injecting the AI consensus directly into registration portals.